Evidence Routing and Supported Residuals for Cooperative LiDAR–Radar Perception

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***Abstract*—Missing light detection and ranging (LiDAR) returns and spurious echoes require different responses before an agent shares features for cooperative perception. We study how to route these observations and constrain radar-derived corrections within a common LiDAR–radar detector. The DEER module learns independent missing/spurious scores from traceable deletion/injection targets and detaches them when routing features, retaining a positive gain for each available modality. The SBR module selects nearby measured radar donors with a null option and adds a residual constrained by observation support and current feature amplitude. Detection supervision learns the correction direction without a feature teacher. Seven V2X-R configurations share one seed, 20 training epochs, and model selection on Normal/Fog validation data; Snow is reserved for final testing. The predetermined joint model AB achieves 49.19 percent Vehicle three-dimensional average precision (AP3D) at an intersection over union (IoU) of 0.7 in Snow, versus 45.67 percent for the internal baseline. Its Fog/Snow mean increases by 1.67 percentage points, while Normal/Fog AP3D decreases by 0.26/0.19 points. A multiplicative type-prior control leads in Normal, Fog, and the weather mean (50.62 versus AB's 50.13 percent). The results support a Snow-specific benefit and identify a weather-dependent tradeoff in residual control.**

***Index Terms*—cooperative perception, LiDAR–radar fusion, adverse weather, evidence routing, bounded residual**

I. Introduction

Cooperative perception combines observations across connected agents to extend their coverage beyond individual viewpoints. Attentive aggregation, heterogeneous interactions, and spatially selective communication improve feature exchange [1]–[3]. The quality of the exchanged features also matters: sharing a locally corrupted representation can propagate measurement errors to the cooperative detector. We examine how an agent should construct that representation from light detection and ranging (LiDAR) and radar under adverse weather.

Adverse weather can remove LiDAR returns through attenuation and add spurious returns through backscatter [4], [5]. Suppressing suspect features addresses clutter but cannot supply missing evidence. Radar can contribute complementary observations, yet its sparse coverage makes an unrestricted correction poorly supported in some regions. The central design problem is therefore to distinguish the evidence used for routing from the measured support and allowable correction amplitude.

L4DR combines multimodal encoding, foreground-aware denoising, and gated fusion [6]; V2X-R uses radar-conditioned denoising diffusion [7]; HRCP couples cross-modal gates with residual enhancement [8]. We examine the relationship between routing and correction control: can separately supervised missing/spurious evidence guide routing while measured radar support and current feature scale constrain correction location and amplitude?

We implement this relationship before a shared AttFuse cooperative backbone. DEER (A) predicts two independent degradation scores from provenance-labeled perturbations and uses detached scores to route features without erasing an available branch. SBR (B) aggregates nearby measured radar donors, permits donor rejection, and bounds its supported residual in current feature units. Detection supervision learns the correction direction; no normal-feature teacher is required. This design separates the semantic evidence guiding a response from the observation support that permits it.

The contributions are:

- An evidence-routing mechanism with independent deletion/injection targets, detached type control, retained modality gains, and a current-amplitude descriptor.
- A task-trained correction mechanism with local measured donors, a null option, a hard support mask, and a channelwise amplitude bound, inserted before unchanged cooperative aggregation.
- Seven budget-matched configurations that test the modules jointly and separately, an ordinary gating/residual control, and amplitude/type-prior alternatives. AB gains 3.52 points in Snow three-dimensional average precision (AP3D) at an intersection over union (IoU) of 0.7; the type-prior control instead leads the Fog/Snow mean, defining the observed benefit's scope.

II. Related Work

A. Cooperative Three-Dimensional Perception

OPV2V introduces attentive intermediate fusion [1], V2X-ViT models heterogeneous inter-agent and spatial interactions [2], and Where2comm selects transmitted regions using spatial confidence [3]. CoAlign handles pose errors through agent–object pose graphs and multiscale fusion [9]. These methods address collaboration or alignment. We fix that stage and change the local multimodal features supplied to it, using the poses and calibration supplied with the dataset.

B. Weather-Aware LiDAR–Radar Fusion

Physical fog and snowfall models distinguish attenuation from added returns [4], [5]. L4DR first filters radar points by foreground probability, exchanges information within shared pillars, and gates intra-modal features using fused features [6]. DEER instead supervises two LiDAR perturbation fractions and retains positive gains for both available branches. V2X-R's multimodal denoising diffusion (MDD) iteratively denoises LiDAR features conditioned on radar and uses a noise-masked reference-feature loss [7]. SBR predicts one local residual directly from the detection objective and magnitude penalties.

HRCP offers a close structural comparison: its BCMG generates sigmoid gates from the opposite modality, and AFE embeds the gated features with original radar features to predict a residual through channel attention [8]. Our distinction lies in explicit provenance supervision and detached type-control paths, followed by donor selection and a measured-support mask with a current-amplitude bound. HRCP additionally changes how the two modalities collaborate; our cooperative stage is shared across controls. These comparisons concern mechanisms; performance is not compared under a common protocol.

C. Reliability Modeling and Controlled Correction

RAF supervises camera reliability and uses calibration-aware local matching with camera, LiDAR, and radar inputs [10]. DEER uses point clouds and predicts controlled deletion/injection fractions rather than detection correctness. SBR separately represents donor rejection and output support: null attention can reduce donor mass, whereas the hard mask prevents correction without a measured radar candidate. Neither constraint alone implies improved detection.

III. Method

A. Problem Definition

For agents $i = 1, \dots, n$, let $P_i^L \in \mathbb{R}^{N_i^L \times 4}$ and $P_i^R \in \mathbb{R}^{N_i^R \times 4}$ be current point clouds. Each point contains coordinates and the source framework's red-channel scalar divided by 255; this scalar is not reinterpreted as calibrated velocity. Given the supplied poses and calibration, the loader projects points once onto the ego grid. Agents within 70 m are retained, up to five including the ego. The output is a set of scored ego-frame Vehicle boxes with seven geometric parameters. RGB pixels, weather-label color channels, provenance IDs, and reference features are excluded from model inputs.

Separate pillar encoders and scatter operations [11] produce $F_i^L, F_i^R \in \mathbb{R}^{64\times200\times704}$. The ten-channel observation tensor Ω contains, for each modality, point count, occupancy, geometric field-of-view (FOV) validity, sensor-relative distance, and a broadcast availability flag $v_i^m \in \{0, 1\}$. Distance is divided by the bird's-eye-view (BEV) range diagonal; counts enter the control encoder through $\log(1 + \cdot)$. Validity includes accepted measured cells at FOV boundaries. Occupancy is computed after geometric filtering, before voxel sampling. Empty cells and unavailable frames are distinct; neither removes a sample or its ground-truth (GT) boxes. Below, $C = 64$, $H = 200$, $W = 704$, and b indexes a grid cell.

B. Overall Architecture and Feature Units

Fig. 1 shows the insertion point between scatter and the common cooperative backbone.

Suppressing the agent index where unambiguous, the shared local path is

$$Z_0 = \Phi_{\text{base}}([F^L, F^R]), \tag{1}$$

$$X = \Phi_{\text{out}}([Z', F^R]), \qquad \widehat{\mathcal{B}} = \mathcal{H}(\mathcal{C}(\{X_i\}_{i=1}^n)), \tag{2}$$

where brackets concatenate channels, Z' is the optionally routed/corrected feature, $\mathcal{C}$ is the shared multiscale backbone with AttFuse, and $\mathcal{H}$ is the detection head. Bias-free pointwise projections Φ_{base} and Φ_{out} map $2C \to C$ and $2C \to 2C$, initialized at $[I, 0]$ and I. They remain trainable in every configuration, including the internal baseline S0. The original radar bypass is common to all comparisons.

For any feature F, let $U(b)$ mark valid occupied cells and sg denote stop-gradient. The implemented per-agent, per-channel root-mean-square (RMS) operator $s = \mathcal{R}(F, U)$ is

$$\begin{aligned} e_c &= \frac{\sum_b U(b)\,\text{sg}(F_c(b))^2}{\max(\sum_b U(b), 1)}, \\ \bar{s} &= \text{clip}\left(\sqrt{C^{-1}\textstyle\sum_c e_c}, \epsilon, M\right), \\ s_c &= \text{clip}(\text{clip}(\sqrt{e_c}, 0.1\bar{s}, 10\bar{s}), \epsilon, M), \end{aligned} \tag{3}$$

with $\epsilon = 10^{-3}$ and $M = 10^3$. This detached operator has no learned parameters, running averages, or normal-reference calibration. Empty support receives a positive numerical floor without being treated as an observation. Only control inputs and correction units are normalized; the detection features and radar bypass retain their learned units.

C. DEER: Missing and Spurious Evidence Routing

1) Independent targets and predictions: On a Normal training slot, the controlled deletion/injection process tracks the source identities of original LiDAR points. For each geometrically filtered cell, let N_{ref}, N_{del}, N_{obs}, and N_{inj} count reference, deleted, observed, and injected points. The targets are

$$y_{\text{miss}} = \frac{N_{\text{del}}}{\max(N_{\text{ref}}, 1)}, \qquad y_{\text{spur}} = \frac{N_{\text{inj}}}{\max(N_{\text{obs}}, 1)}. \tag{4}$$

Their validity masks are $U_{\text{miss}} = \mathbf{1}[N_{\text{ref}} > 0]$ and $U_{\text{spur}} = \mathbf{1}[N_{\text{obs}} > 0]$. Thus deletion and injection may coexist, and unknown cells are excluded rather than assigned a negative target. The numerators come from recorded source events, not differences in total point counts. Unperturbed Normal observations provide zero targets on their valid occupied cells. Fog type targets remain masked because the point-origin semantics of these clouds have not been verified. Normal raw points support label construction during training; they are never encoded as teacher features.

Compute $s_L = \mathcal{R}(F^L, U_L)$ and $s_R = \mathcal{R}(F^R, U_R)$ on each modality's valid measured occupancy, and let Ω collect the ten

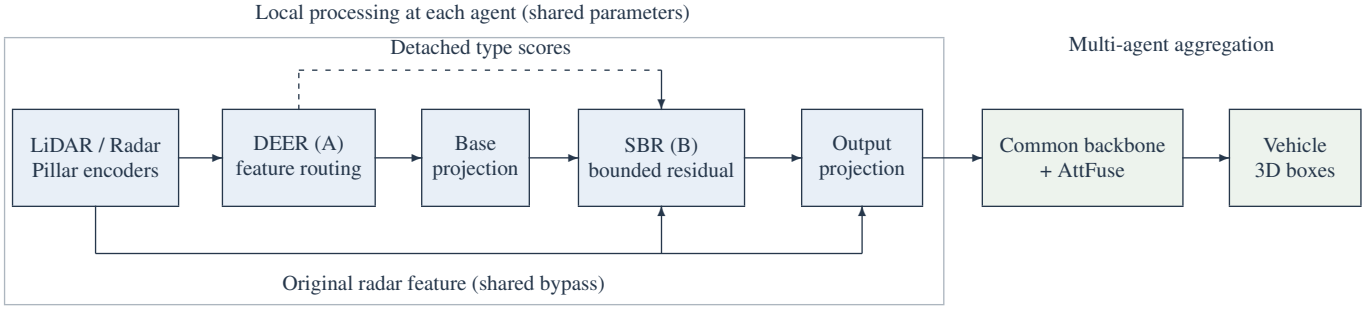


Fig. 1. Local LiDAR–radar processing (blue) before common cooperative aggregation and detection (green). Solid arrows carry features; the dashed path carries detached missing/spurious scores. SBR also reads current observation support. All configurations share the original radar bypass; type targets are used only in the training loss.

current-observation/availability maps. The control representation is

$$a_{\text{amp}} = [\log(1+s_L), \log(1+s_R)],$$
$$E = f_E\big(W_C[F^L/s_L, F^R/s_R, \Omega] + W_A a_{\text{amp}}\big). \quad (5)$$

Here W_C projects $2C+10$ channels to 16 with bias. Bias-free W_A maps $2C$ amplitude values to 16 at spatial size 1×1 and broadcasts them, adding 2,048 parameters. The encoder applies SiLU, a 16-channel depthwise 3×3 convolution, SiLU, a $16\to16$ pointwise convolution, and SiLU; all spatial sizes are preserved. Separate pointwise type and route heads each output two channels. Independent sigmoids produce d_{miss} and d_{spur}.

2) Detached control and retained gains: Learned route logits $h_L(E), h_R(E)$ receive type biases:

$$\ell_L = h_L(E) - \text{sg}(d_{\text{spur}}), \qquad \ell_R = h_R(E) + \text{sg}(d_{\text{miss}}). \quad (6)$$

For an available modality, define

$$a_m = \frac{v^m \exp(\ell_m)}{\sum_{k\in\{L,R\}} v^k \exp(\ell_k)},$$
$$g_m = v^m[1+\alpha(2a_m-1)], \quad (7)$$

where $\alpha = 0.5$. If both modalities are unavailable, weights and gains are set to zero. The routed representation is

$$Z_A = \Phi_{\text{base}}([g_L F^L, g_R F^R]). \quad (8)$$

An available branch retains gain in $[0.5, 1.5]$; an unavailable branch stays zero. Zero type-head weights/biases and compensated route-head biases initialize both gains to one when both modalities are available, or to 1.5 for a sole available branch. Stop-gradient blocks detection gradients through the type outputs; detection and auxiliary gradients still meet in the shared encoder.

The descriptor in (5) restores amplitude information suppressed by control normalization. N1 tests an active alternative: occupied channel mean divided by the same RMS, with identical projection size and initialization.

D. SBR: Observed Support and Bounded Residual Repair

1) Local donors and rejection: SBR receives $Z = Z_A$ when A is enabled and $Z = Z_0$ otherwise. At cell b, the donor set $\mathcal{K}(b)$ contains radar cells within 0.8 m whose occupancy, FOV validity, and frame availability indicate a real measurement. The 0.4 m grid gives 13 disk offsets inside a 5×5 bounding window. Convolutional activation alone cannot admit a donor.

The feature scale $s = \mathcal{R}(Z, U_L \vee U_R)$ uses the union of available valid occupancies. If the unguarded overall RMS of an agent's Z is at most ϵ and real radar exists, s falls back to that radar's current occupied RMS s_R. This accommodates a zero-initialized LiDAR-led base feature without limiting the radar correction to a numerical floor. Define $\widehat{Z} = Z/s$ and $\widehat{R} = F^R/s_R$.

Bias-free pointwise projections produce $q = Q\widehat{Z}$ and $k = K\widehat{R}$ in eight channels; $h_\varnothing$ is a zero-initialized affine $C\to1$ projection. The unit-ball map $\nu(x) = x/\max(\|x\|_2, 1)$ gives real-donor scores

$$t_u(b) = 2\nu(q(b))^\top \nu(k(u)), \quad u\in\mathcal{K}(b),$$
$$t_\varnothing(b) = h_\varnothing(\widehat{Z}(b)) + \log\max(|\mathcal{K}(b)|, 1). \quad (9)$$

Real scores lie in $[-2, 2]$. A softmax over the valid candidates and always-valid null donor yields π. The normalized donor feature and hard support mask are

$$\widehat{D}(b) = \sum_{u\in\mathcal{K}(b)} \pi_u(b)\widehat{R}(u), \quad S_R(b) = \mathbf{1}[|\mathcal{K}(b)| > 0]. \quad (10)$$

The null feature is zero; real weights are not renormalized after removing its mass. The count correction in (9) gives null mass $1/2$ when all uncorrected logits are zero and candidates exist. With no candidate, null mass is one and $\widehat{D} = 0$.

2) Correction and explicit bound: Let h_A indicate that A supplies scores, and let o_L be LiDAR occupancy. SBR's residual network reads

$$r = f_B([\widehat{Z}, \widehat{D}, \text{sg}(d_{\text{miss}}), \text{sg}(d_{\text{spur}}), o_L, S_R, h_A]),$$
$$\Delta_c(b) = S_R(b)\rho s_c \tanh(r_c(b)), \quad Z' = Z + \Delta, \quad (11)$$

where $\rho = 0.25$. A pointwise projection maps $2C+5$ channels to 16, followed by SiLU, a depthwise 3×3 convolution, SiLU,

and a $16 \to C$ pointwise output. Without A, type scores and h_A are zero. Without B, $Z' = Z$.

The output obeys $\Delta = 0$ when $S_R = 0$ and $|\Delta_c(b)| \le \rho s_c$. A zero-initialized final layer gives an initial identity update. These bounds apply to the added correction. Null attention reduces donor evidence but cannot by itself force a zero residual when real support exists.

N2 tests an additional amplitude prior by multiplying (11) by

$$G_A = 0.25 + 0.75 \max(\mathrm{sg}(d_{\mathrm{miss}}), \mathrm{sg}(d_{\mathrm{spur}})). \tag{12}$$

AB uses type scores as inputs only; N2 additionally applies this multiplier.

E. Joint Training and Losses

The shared detection loss is focal classification plus twice smooth-L_1 box regression, with focal parameters 0.25 and 2 and smooth-L_1 transition $1/9$. Regression sums the seven anchor-relative codes, using a sine yaw difference. Each sample's classification and regression sums are divided by its positive-anchor count (at least one), then averaged across cooperative samples. Regression uses only positives; classification ignores anchors marked neither positive nor negative.

For a mask U, $\langle V \rangle_U$ sums masked entries over all agents, cells, and applicable channels in the microbatch, divided by the broadcast mask sum clamped to one. With n_i marking an unperturbed Normal agent slot, the auxiliary losses are

$$\begin{aligned}
\mathcal{L}_{\mathrm{type}} &= \sum_{t \in \{\mathrm{miss},\mathrm{spur}\}} \langle \mathrm{BCE}(d_t, y_t) \rangle_{U_t}, \\
\mathcal{L}_{\mathrm{size}} &= \langle (\Delta / \mathrm{sg}(s))^2 \rangle_{S_R}, \qquad \mathcal{L}_{\mathrm{id}} = \langle |\Delta / \mathrm{sg}(s)| \rangle_{n_i S_R}, \\
\mathcal{L} &= \mathcal{L}_{\mathrm{det}} + 0.1\mathcal{L}_{\mathrm{type}} + 0.001\mathcal{L}_{\mathrm{size}} + 0.01\mathcal{L}_{\mathrm{id}}.
\end{aligned} \tag{13}$$

Binary cross-entropy is evaluated on the corresponding logits. Empty masks give differentiable zero; inactive modules contribute zero for their associated terms. The loss weights remain fixed without warmup. Detection learns the signed correction; the squared size penalty acts on supported cells, and the absolute identity penalty acts only on unperturbed Normal support. No reconstruction or teacher loss is used.

Training constructs provenance targets outside the model, computes the current-input features and predictions, and backpropagates (13). Inference follows the same local feature path with fixed weights and batch-normalization (BN) statistics, without perturbations or auxiliary targets, then decodes and suppresses boxes. Current RMS scales are recomputed from each observation in both modes.

IV. Experiments

A. Experimental Setup

1) Data and weather: We use the simulated cooperative V2X-R dataset [7]. Table I reports the actual inventory: the training split contains 8,083 samples, one fewer than reported in the dataset description. Agent sensor frames are not independent cooperative samples.

Paired Normal/Fog/Snow clouds are supplied by the dataset authors; radar is unchanged. The suffixes `_fog_0.060` and `_snow_2.5_2.0` identify weather versions without assigning physical meanings to their tokens. In every configuration, each training agent uses Fog with probability $3/7$ and Normal otherwise, deterministically indexed by seed, epoch, sample, and agent. Snow is excluded from training, validation, tuning, and model selection.

Normal slots receive the same provenance augmentation in all groups. With probability 0.5, original points are independently deleted with probability 0.2 and $\lceil 0.05N \rceil$ injected points copy randomly sampled originals, including their scalar field. Offsets have isotropic directions and uniformly sampled radii in $[0.075, 0.3]$ m. Injected points are filtered by the same sensor self-mask, FOV, and range rules. Realized fractions therefore vary. This augmentation supplies traceable labels, not a physical weather simulation.

2) Configurations and implementation: Table II specifies all seven completed groups. S0 adapts pillar encoding and AttFuse within the common path of Fig. 1; it is an internal baseline. No published method is re-evaluated under this protocol. SUP retains the observation encoder, type auxiliary loss, augmentation, and residual regularizers, but removes type control, local donor attention, hard output support, and the $\tanh$ bound. Its donor is the current normalized radar feature; penalties still use the common support mask. SUP changes several paths and only approximately matches capacity.

The pillar size is $0.4 \times 0.4 \times 4$ m over $[-140.8, 140.8) \times [-40, 40) \times [-3, 1)$ m, with at most 32 points per pillar and 32,000/70,000 pillars per modality in training/evaluation. Backbone stages have 128/256/512 channels; each starts with a stride-2 3×3 convolution followed by 3/5/8 stride-1 convolutions. AttFuse aggregates agents at each scale; upsampling yields 128 channels per scale and a concatenated 384-channel input to the pointwise classification/regression heads. Vehicle anchors have length/width/height $3.9/1.6/1.56$ m, headings $0/90$ degrees, and output stride 2. Assignment uses continuous stand-up BEV IoU: positives at 0.6 or above, negatives below 0.45, with a positive-overlap best-anchor assignment per GT. Intermediate anchors are ignored. All groups share these rules.

Geometric augmentation uses an x-axis world flip, rotation in $[-\pi/4, \pi/4]$, and scaling in $[0.95, 1.05]$. Optional local branches use activation checkpointing during training. This recomputes activations for backpropagation and leaves the inference graph unchanged.

3) Training and selection: Every group completes 20 epochs and 20,200 optimizer updates with seed 17. FP32 microbatch 4 and two accumulation steps give effective batch

TABLE I
Actual Split Inventory and Fixed Vehicle Ground Truth

Split	Coop. samples	Sensor frames	Fixed GT
Train	8,083	24,088	69,757
Validation	829	3,041	6,414
Test	3,166	10,598	30,296

TABLE II
COMPLETED CONFIGURATIONS; SHARED PROTOCOL IN SEC. IV-A

Group	Local mechanism or controlled change
S0	Common projections, pillar encoders, radar bypass, backbone, and head; A/B disabled.
A	DEER only.
B	SBR only; no A predictions or type inputs.
AB	Predetermined joint model; detached types enter B as inputs without an extra multiplier.
SUP	Ordinary gating/residual control (Sec. IV-A).
N1	AB with occupied mean/RMS instead of the RMS amplitude descriptor; same projection size and initialization.
N2	AB with the multiplicative type prior in (12); all other settings unchanged.

8; the incomplete effective batch is dropped each epoch. Independent single-GPU experiments are scheduled across five RTX 2080 Ti GPUs. S0 resumes an audited online checkpoint at update 1,000; its total budget remains identical. The other groups use the same initialization for shared parameters. Adam uses learning rate 0.002, weight decay 10^{-4}, and $\epsilon = 10^{-10}$; the rate is multiplied by 0.1 at epochs 10 and 15. Loss weights are fixed as in (13).

After each epoch, validation uses all 829 samples under both Normal and Fog conditions. Before evaluation, BN buffers are reset and refitted using cumulative averaging over eight microbatch-4 passes through 32 fixed training samples. Uniform midpoint sampling uses the epoch-0 Normal/Fog/provenance stream without geometric augmentation. Parameters remain fixed, other modules stay in evaluation mode, and only BN statistics update. BN has $\epsilon = 0.001$ and online momentum 0.01. Every group uses this refit protocol.

Selection maximizes mean Normal/Fog validation AP3D at IoU 0.7; ties within 10^{-6} retain the earlier epoch. N1/N2 select epoch 18 and the others epoch 20. Each selected model uses identical weights and frozen BN for all three test conditions, without weather-specific refitting or reselection.

4) Scoring: Vehicle AP3D and BEV average precision (APBEV) are computed using 3D and BEV IoU, respectively, at thresholds 0.3/0.5/0.7. Predictions are globally sorted by confidence and matched one-to-one to an unmatched GT in the same sample; AP uses VOC2010 all-point interpolation. The fixed GT set retains original boxes whose ego-frame centers satisfy $x \in [0, 140)$ m and $y \in [-40, 40)$ m and a common camera-geometric FOV. Calibration and image-size metadata define the FOV without RGB pixels. No point-count rule or GT cap removes boxes; zero-target samples remain.

Postprocessing fixes the confidence floor at 0.001, rotated BEV non-maximum suppression (NMS) IoU at 0.15, and maximum detections at 500. Each of the 21 tests covers all 3,166 samples and 30,296 fixed GT boxes. The adverse-weather summary is

$$M_{\text{weather}} = \tfrac{1}{2}\left(\text{AP3D}_{0.7}^{\text{Fog}} + \text{AP3D}_{0.7}^{\text{Snow}}\right). \tag{14}$$

This one-class, two-weather mean differs from the Normal/Fog validation selection score. AP is a percentage (higher is better); differences are in percentage points and are computed from unrounded CSV values before rounding to two decimals.

TABLE III
TEST VEHICLE AP3D AT IOU 0.7 (%; HIGHER IS BETTER). M_{weather} IS DEFINED IN (14); ΔM IS ITS CHANGE FROM S0 IN PERCENTAGE POINTS. BOLD INDICATES COLUMN MAXIMA.

Group	Normal	Fog	Snow	M_{weather}	ΔM
S0	65.46	51.25	45.67	48.46	+0.00
A	64.67	51.60	46.22	48.91	+0.45
B	65.18	51.41	44.72	48.06	-0.40
AB	65.19	51.07	**49.19**	50.13	+1.67
SUP	64.84	51.36	46.24	48.80	+0.33
N1	64.37	51.03	48.69	49.86	+1.40
N2	**65.87**	**52.52**	48.73	**50.62**	+2.16

TABLE IV
DERIVED FOUR-CELL EFFECTS AT IOU 0.7 (PERCENTAGE POINTS). I IS DEFINED IN (15); VALUES ARE DERIVED FROM THE RUNS IN TABLE III.

Condition	A−S0	B−S0	AB−S0	I
Normal	-0.79	-0.28	-0.26	+0.81
Fog	+0.35	+0.16	-0.19	-0.69
Snow	+0.54	-0.96	+3.52	+3.93
M_{weather}	+0.45	-0.40	+1.67	+1.62

B. Detection Across Weather Conditions

Table III shows a weather-dependent benefit. AB has the highest Snow AP3D at IoU 0.7, 49.19 versus S0's 45.67 percent, gaining 3.52 points. Its weather mean increases by 1.67 points, but Normal/Fog AP3D decreases by 0.26/0.19 points. N2 leads Normal, Fog, and the weather mean, with a mean gain of 2.16 points over S0. Relative to N2, AB gains 0.46 in Snow but loses 0.67/1.45 points in Normal/Fog and 0.49 on the mean. AB remains the predetermined joint model throughout reporting.

AB–S0 Snow gains grow from +0.24 at IoU 0.3 to +0.75 at 0.5 and +3.52 at 0.7. The corresponding changes in Normal are $-0.20/-0.14/-0.26$ and Fog changes $-0.08/+0.24/-0.19$ at these thresholds. The stricter-overlap gain is consistent with a localization-related benefit, but AP also reflects ranking and false positives; aggregate AP cannot identify feature repair as its cause.

C. Module Interaction and Design Controls

1) Separate and joint modules: Table IV derives effects from the S0/A/B/AB rows in Table III. A gains 0.45 points in the weather mean, while B loses 0.40, including a 0.96-point Snow decrease. The joint design outperforms A/B by 2.98/4.48 points in Snow and 1.22/2.07 on the mean. Define the descriptive interaction

$$I = \text{AP}_{AB} - \text{AP}_{A} - \text{AP}_{B} + \text{AP}_{S0}. \tag{15}$$

Its positive Snow value (+3.93) and negative Fog value (-0.69) show that the positive non-additivity occurs in Snow. With nonlinear AP and one seed, this contrast does not establish causal complementarity or statistical significance. B's loss also shows that the structural constraints alone do not ensure improved detection.

TABLE V
AUXILIARY TEST AP (%; HIGHER IS BETTER). LEFT: SNOW AP3D AT IOU 0.3/0.5; RIGHT: APBEV AT IOU 0.7 FOR EACH WEATHER CONDITION. BOLD INDICATES COLUMN MAXIMA.

	Snow AP3D		APBEV at IoU 0.7		
Group	IoU 0.3	IoU 0.5	Normal	Fog	Snow
S0	82.64	76.61	77.59	62.99	63.86
A	83.17	77.18	77.20	63.61	64.87
B	82.73	76.14	76.71	62.62	62.63
AB	82.87	77.37	77.40	63.74	65.42
SUP	**83.78**	**77.81**	**77.61**	63.36	65.20
N1	83.52	77.78	77.26	63.07	65.11
N2	83.37	77.57	77.48	**63.98**	**65.56**

2) Ordinary control SUP: AB exceeds SUP by 2.96 points in Snow and 1.33 in the weather mean, with a 0.35-point Normal gain and a 0.29-point Fog loss. Shared augmentation and type supervision therefore do not reproduce AB's result with ordinary gating/residual control. Because SUP changes multiple paths, this comparison cannot isolate any single mask, bound, or control connection.

3) Amplitude descriptor N1: AB exceeds the equally parameterized occupied-mean/RMS alternative by 0.82/0.04/0.50 points in Normal/Fog/Snow and 0.27 in the weather mean. This result favors the RMS descriptor in this run; the near-zero Fog difference and unknown training variability limit the conclusion.

4) Type-prior control N2: N2 retains the type inputs, local donors, null option, support mask, and bound, adding only (12). Its higher weather mean contradicts a general advantage for input-only type conditioning, while AB retains the 0.46-point Snow advantage. The comparison identifies a tradeoff between weather conditions; it does not establish whether the multiplier over-suppresses useful corrections or why it helps Fog.

D. Metric and Selection Sensitivity

Table V shows that the Snow lead is specific to strict 3D overlap. N2 leads Snow APBEV at IoU 0.7 (65.56 versus AB's 65.42 and S0's 63.86). SUP leads Snow AP3D at 0.3/0.5 (83.78/77.81 versus AB's 82.87/77.37). The full six-metric, three-weather results remain in the accompanying CSV.

Normal/Fog validation means are 49.54 for AB and 49.15 for S0 (+0.39); SUP leads at 51.22. On test, the same Normal/Fog mean is 58.13 versus 58.36 (-0.23), whereas the Fog/Snow mean ranks N2 first and AB second. These split/weather differences explain why the rankings answer different questions, while the rankings alone do not establish overfitting. All selections and scoring settings remain frozen.

V. LIMITATIONS

The evidence covers one seed, one simulated dataset, and 20 epochs. There are no variance estimates or significance tests, and equal budgets do not imply equal convergence. Small differences, such as the 0.27-point mean difference between AB and N1 and the 0.46-point Snow AP3D difference between AB and N2, require validation across repeated runs. Evaluation on the supplied weather versions does not establish transfer to real roads, other weather severities, or joint radar degradation. Provenance targets describe controlled perturbations, and a finite donor neighborhood may omit useful returns or include clutter.

Internal controls support comparisons within this detector; same-protocol external baselines and finer mechanism ablations are unavailable. The results do not establish state-of-the-art performance or module-level causal explanations. No matched training/inference latency, memory, communication, or energy study is available; independent job scheduling supplies no deployment-efficiency evidence.

VI. CONCLUSION

Separating degradation evidence from correction support provides a concrete way to organize local LiDAR–radar processing before cooperation. DEER supplies detached missing/spurious control while retaining available branches; SBR adds a task-trained residual only within measured radar support and a current-amplitude bound. In the single-seed V2X-R evaluation, AB gains 3.52 points in Snow AP3D at IoU 0.7 and 1.67 on the Fog/Snow mean, with Normal/Fog losses. The multiplicative-prior control achieves the higher mean (50.62 versus 50.13 percent). Thus the demonstrated benefit concerns Snow detection under this protocol, while the choice of type conditioning for the residual remains weather-dependent.

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